

**Evaluating Haptic Feedback Modalities for the
Robotiq 2F-85 Adaptive Gripper using
Stress-Deformation Tactile Sensors**

Bachelor's Thesis

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Abstract

Teleoperation of robotic systems, such as pick-and-place tasks, often suffers from a lack of haptic feedback, forcing operators to rely entirely on visual cues. This reliance can lead to the application of excessive force, damaging delicate objects, or insufficient force, leading to slippage. This thesis investigates a novel bilateral teleoperation framework that integrates a Robotiq 2F-85 Adaptive Gripper equipped with stress-deformation-based tactile arrays. The tactile data generated during grasping is translated in real-time into spatial vibrotactile feedback delivered to the operator via a customized 5-channel ERM multi-actuator array driven by an ESP32-C6 microcontroller. Through a series of quantitative latency benchmarking tests and qualitative participant surveys, this study evaluates the efficacy of different vibrotactile mapping modalities. Results aim to demonstrate whether stress-deformation feedback significantly improves perception and force-regulation during the manipulation of compliant versus rigid objects.

Chapter 1

Introduction

1.1 Background and Motivation

In modern industrial and remote-assistance applications, robotic teleoperation serves as a critical bridge between human cognitive decision-making and mechanical execution in hazardous or inaccessible environments. However, while visual feedback systems have advanced considerably, the transmission of the sense of touch—haptics—remains a significant bottleneck.

When a human user operates a robotic manipulator, such as the widely used Robotiq 2F-85 Adaptive Gripper, they must infer the grip force and the structural integrity of the target object purely through a multi-dimensional video feed. This visual-only paradigm is fundamentally flawed when dealing with non-rigid, compliant, or fragile objects.

1.2 Problem Statement

Basic industrial gripper units, even those capable of adaptive grasping geometries, lack localized high-resolution tactile sensors capable of discriminating normal and shear forces dynamically. Furthermore, safely relaying highly dense tactile data back to a human operator requires translating mechanical stress matrices into discernible somatosensory stimuli.

1.3 Thesis Objectives

This research aims to answer the following core objectives:

1. To integrate a high-resolution, stress-deformation-based tactile sensor suite into the fingers of a Robotiq 2F-85 Adaptive Gripper.
2. To develop a robust, low-latency firmware mapping protocol using an ESP32-C6 host to trigger an array of Eccentric Rotating Mass (ERM) vibrotactile actuators.
3. To systematically evaluate participant performance and subjective qualitative experience when grasping delicate objects using different haptic feedback mappings.

1.4 Thesis Structure

This thesis is structured as follows: Chapter 2 reviews the state of the art in tactile sensing and teleoperated haptics. Chapter 3 details the system architecture, detailing both the Modbus RTU communication with the Robotiq 2F-85 and the ESP32-C6 multi-actuator setup. Chapter 4 outlines the experimental survey methodology. Chapter 5 presents the quantitative and qualitative results. Finally, Chapters 6 and 7 provide a discussion and conclusion of the findings.

Chapter 2

Literature Review

2.1 Robotic Grasping and the Robotiq 2F-85

The Robotiq 2F-85 Adaptive Gripper represents an industry standard in collaborative robotics. With its patented underactuated finger design, it passively adapts to the shape of the target object. However, its built-in feedback is generally limited to simple collision detection and binary grasp status derived from motor current thresholds. ...

[Draft Note: Expand this section discussing limits of motor-current based force estimation.]

2.2 Stress-Deformation Tactile Sensing

To bridge the gap between gross motor feedback and fine tactile discrimination, researchers have increasingly turned to elastomeric optical sensors and stress-deformation arrays. These sensors map the continuous deformation of a soft skin into a digital pressure topology. ...

2.3 Vibrotactile Feedback Mechanisms

Among various haptic rendering methods (kinesthetic force-feedback, electro-tactile, thermal), vibrotactile actuation via ERM (Eccentric Rotating Mass) and LRA

(Linear Resonant Actuator) motors remains the most ubiquitous due to its cost-effectiveness, scalability, and ease of integration into wearable interfaces. ...

Chapter 3

System Architecture

3.1 Overall Framework Design

The teleoperation loop consists of a host PC bridging the robotic end-effector and the human operator array.

3.2 Robotiq 2F-85 and Tactile Integration

The primary manipulation hardware employs the Robotiq 2F-85. Communication is established via Modbus RTU over RS-485 utilizing the `pyRobotiqGripper` Python library, enabling dynamic setpoints for position, speed, and force.

3.2.1 Sensor Integration and Calibration

The stress-deformation tactile sensors are mechanically affixed to the internal, parallel-facing phalanges of the gripper. The raw electrical response from the piezoelectric matrix is filtered and calibrated to return an array of continuous pressure floats ranging from 0.0 to 1.0.

3.3 ESP32-C6 Haptic Driver

On the operator side, localized pressure vectors are mapped to five distinct ERM motors. The ESP32-C6 utilizes a customized MicroPython firmware running a dedicated `stream_mode` event loop. It parses 20-byte binary structures over serial

via standard input, translating incoming arrays of 32-bit floats into 10-bit PWM duty cycles (0–1023) at an optimized frequency of 200 Hz.

3.3.1 Motor Driving Circuitry

To safely drive the ERMs, NSLEEP and Motor Enable (EN) logic pins are configured to strictly govern the H-bridge drivers, ensuring fail-safes are triggered if the serial watchdog times out.

Chapter 4

Experimental Methodology

4.1 Survey and Task Design

To evaluate the efficacy of the proposed system, an Institutional Review Board (IRB) approved survey was developed. Participants ($N = \dots$) were tasked with remotely manipulating three distinct classes of objects:

- Rigid Objects (e.g., steel blocks)
- Compliant Objects (e.g., silicone spheres)
- Fragile Objects (e.g., eggshells or delicate plastics)

4.2 Feedback Modalities Tested

Participants completed the tasks under three conditions:

1. **Visual Only:** No haptic feedback is provided.
2. **Binary Collision Feedback:** A standard burst of vibration is triggered across all motors purely indicating that object contact has been made.
3. **Proportional Tactile Mapping:** The intensity of each of the 5 ERM motors corresponds linearly to the specific stress-deformation localized within the tactile sensor quadrants.

4.3 Data Collection

Quantitative data logged by the host PC include grasp completion time, slippage events, and occurrences of excessive crushing force. Post-task qualitative surveys measure perceived realism, task confidence, and cognitive load utilizing a 7-point Likert scale.

Chapter 5

Results and Analysis

5.1 Quantitative Performance Metrics

[Draft Note: Insert statistical comparisons of grasp efficiency here. You may use ANOVA tests to compare compliance failure rates between the three modalities.]

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5.2 Qualitative Survey Results

[Draft Note: Insert Likert scale distribution charts and discuss user feedback regarding the naturalness of the proportional tactile mapping versus the binary mapping.]

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Chapter 6

Discussion

6.1 Interpretation of the Findings

The results highlight the critical necessity of spatial haptic feedback in the manipulation of delicate or compliant objects. While binary collision feedback is sufficient for strict pick-and-place routines involving rigid geometries, it utterly fails to communicate the progressive yield of softer structures.

6.2 Limitations

One persistent limitation observed during the trials was the inherent spin-up and spin-down mechanical latency of ERM motors compared to piezoelectric or LRA alternatives. This delay introduced a slight phase-lag during rapid, impulsive contact events.

Chapter 7

Conclusion

This thesis successfully engineered and evaluated an end-to-end tactile-haptic bilateral teleoperation system. By integrating stress-deformation sensors into the Robotiq 2F-85 Adaptive Gripper and rendering the resultant data via an ESP32-C6 multi-channel ERM array, this study demonstrated that high-resolution proportional feedback significantly reduces error rates in delicate pick-and-place operations.

Future work will focus on substituting the ERM actuators with variable-frequency LRAs to allow for the rendering of high-frequency texture data, independent of grip force amplitude.

Bibliography

- [1] Robotiq Inc. *2F-85 and 2F-140 Adaptive Grippers Instruction Manual*. 2023.
- [2] K. J. Kuchenbecker, J. Fiene, and G. Niemeyer. "Improving contact realism through event-based haptic feedback." *IEEE Transactions on Visualization and Computer Graphics*, 2006.